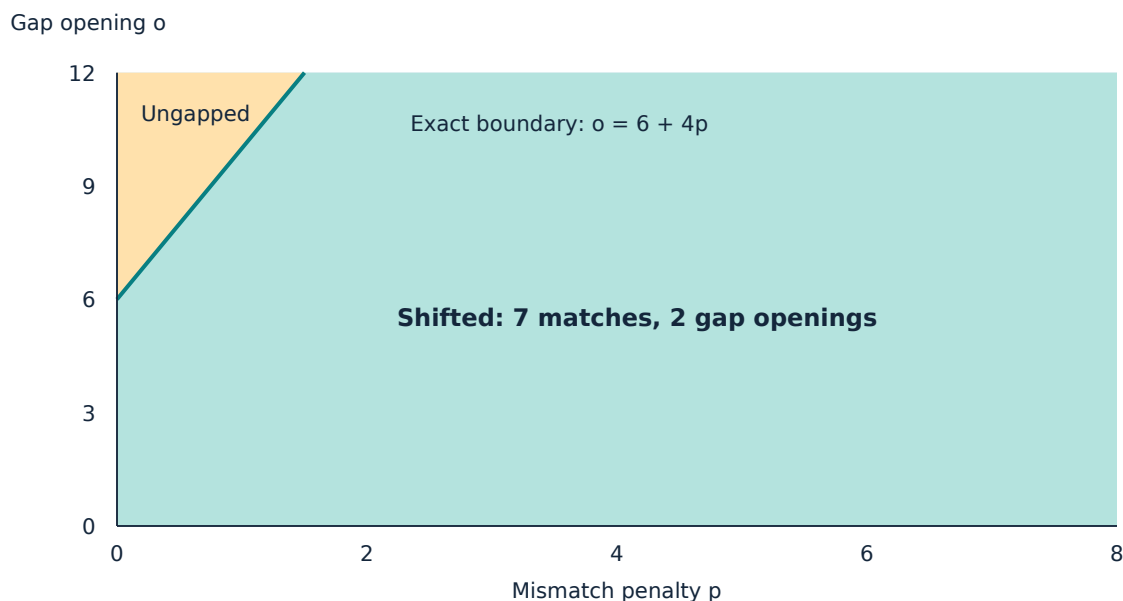


Polynomial exploration of BLAST evidence

Exact alignment populations, scoring geometry, conserved motifs and finite state semantics



3.77 billion operation paths represented by 1,245 feature summaries in the retained twenty-character scaling trial.

Research software monograph
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Purpose and the implemented result

This extension turns sequence-search evidence into explicit, queryable finite mathematical objects. BLAST remains the candidate-discovery engine. PerfectPower imports its tracebacks, binds them to source sequences, constructs complete alignment families for small pairs, computes exact regions of scoring preference, and connects those objects to existing polynomial, state-equivalence, transport, diagnostic, Gamma and graph machinery. The result includes a standard-library Python interface, a command-line interface, public sequence examples, retained BLAST output, regenerated mathematical receipts, Lean modules, and a self-contained interactive HTML atlas.

The implementation has two complementary scales. Scalar affine alignment examines every dynamic-programming cell of a supplied pair and can handle the retained full insulin transcripts. Polynomial alignment retains every admissible feature summary and its multiplicity; this is a more expensive operation and is deliberately limited to smaller sequence pairs. A user can therefore search broadly with BLAST, inspect selected full pairs with scalar alignment, and investigate particular bounded regions with the richer generating-polynomial engine. There is no claim that the polynomial population algorithm is suitable for exhaustive whole-genome alignment.

The retained public example compares human insulin transcript NM_000207.3 with the predicted *Dromiciops gliroides* insulin transcript XM_043971863.1. The downloaded FASTA records have lengths 465 and 483. NCBI BLAST+ 2.12.0, run locally with `blastn`, word size 7 and nucleotide low-complexity masking disabled, returns 17 HSPs. Every imported traceback reproduces its retained source interval. Under match reward 2, mismatch penalty 3, gap opening 5 and gap extension 2, the leading HSP has raw score 174 and agrees with the exhaustive scalar local optimum in this supplied pair. This is a pair-level consistency result, not a database-wide sensitivity or performance result.

The mathematical demonstration uses the illustrative DNA words ACGTACGT and CGTACGTA. Their declared global alignment model has 2,599 operation paths and 76 feature summaries. The exact two-parameter map contains two regions of positive area. An ungapped alignment with eight mismatches competes with a shifted alignment containing seven matches and two one-residue gaps. This example is useful because its ambiguity is visible and its entire finite family is small enough to inspect.

The mathematical object: an alignment population

An admissible global path consumes both finite sequences. A diagonal operation consumes one character from each; a deletion consumes a query character; an insertion consumes a subject character. The model forbids an insertion immediately following a deletion, and vice versa, without an intervening diagonal operation. This choice eliminates immediately opposing gaps from the declared path language. Leading and trailing gaps remain permitted. Characters are compared literally after conversion to uppercase. Ambiguous nucleotide letters are not expanded into sets of possible bases, and U is not silently identified with T.

Each path has the integer feature vector (M,U,G,L), counting identical paired characters, unequal paired characters, gap openings and gap residues. A run of k gap residues contributes one opening and k residues. Consequently the raw score is $rM - pU - oG - eL$. This uses BLAST's open-plus-length-times-extension convention. The common alternative convention that charges extension only after the first gap character

would give different numbers and is not used here.

The generating polynomial is $Z(x,y,z,w) = \sum \text{over admissible paths of } x^M y^U z^G w^L$. Paths with the same feature vector contribute to the same coefficient. A coefficient is therefore a count of operation paths, rather than a count of distinct rendered biological explanations after an unspecified equivalence relation. Repeated letters can make different operation paths look similar; their multiplicity is retained rather than silently discarded.

The implementation computes suffix populations on a finite directed acyclic graph. At a state consisting of query position, subject position and previous operation, each legal outgoing operation shifts every descendant exponent vector by its own feature increment. Alternative paths add coefficients. The terminal state contributes the constant polynomial one. The original state thus contains the complete finite feature distribution for the stated model. This is the sum-product recurrence in a polynomial semiring, carried out with exact integer counts.

The same suffix tables permit rank and selection within a feature fibre. To select the k th path, the algorithm compares k with the number of completions under each outgoing operation in a fixed order. Entire groups of paths can be skipped without listing them. Ranking runs the reverse calculation. Independent recursive enumeration checks both operations on small sequence pairs, including repeated characters, empty sequences and tied score settings.

No feature summary is silently pruned. Limits on retained terms and transition work raise a `WorkLimit` exception instead of returning a misleading complete result. Input sequences are limited to 96 characters each and 128 characters combined for this family compiler. These are implementation budgets, not mathematical restrictions on the definition. The existing sparse PSG polynomial adapter has its own degree and coefficient budgets, so an alignment-family object can exist even when conversion into that narrower polynomial representation is refused.

Scoring geometry and stability of explanations

A signed score vector assigns a linear functional to the feature support. Every feature f is optimal exactly when $\text{dot}(f-g, \text{weights})$ is nonnegative for every competing feature g . The implementation exports these necessary and sufficient inequalities in all four score coordinates. It does not label every support point as a convex-hull vertex: some associated regions can be empty, and some are lower-dimensional tie regions. Feasibility and redundancy are explicitly left unevaluated in that full four-coordinate description.

For one-dimensional parameter paths, the software computes closed optimality intervals by intersecting exact rational inequalities. For two-dimensional parameter planes, it clips a declared rectangular domain against each competing halfspace. The resulting polygon vertices are rational numbers. Degenerate line and point regions are retained, so boundaries do not erase competing explanations. These algorithms work directly with the complete supplied support and do not require floating-point convex-hull decisions.

In the shifted-word example, fix match reward 2 and gap extension 1. Let p be mismatch penalty and o gap-opening penalty. The ungapped explanation has score $-8p$. The shifted explanation has score $12-2o$. Their equality is $o = 6+4p$. Within p between 0 and 8 and o between 0 and 12, the ungapped region has area $9/2$ and the shifted region has area $183/2$. Their areas sum to the rectangle's area 96; the shared boundary represents a tie. The claim is conditional on these illustrative sequences, this path model and this parameter rectangle.

The robustness calculation gives another exact interpretation. For a winning summary f , its score margin over competitor g divided by the L1 norm of $f-g$ is the distance to that competitor's boundary under independent L-infinity perturbations of the four signed weights. Taking the minimum gives the closed radius within which f remains weakly optimal. A tied competitor gives radius zero. If the supplied support contains only one point, no competing boundary exists and the result is represented without a finite radius.

This radius concerns stability of a mathematical ranking. It does not measure biological confidence, account for an incomplete candidate search, or certify that the scoring parameters are scientifically appropriate. It also permits perturbations outside a biologically chosen sign domain unless the caller imposes additional restrictions. The bounded integer-grid interface provides such explicit restrictions through inclusive parameter boxes, rational linear inequalities and modular filters.

Conserved motifs as genuine path constraints

A condition on feature totals cannot tell whether a biologically interesting motif survives intact. Two alignments can have identical counts while placing their mismatches in different locations. The conserved-motif compiler therefore adds a finite automaton to the alignment DAG. It tracks the longest relevant suffix of a contiguous run of identical diagonal columns and a bit mask recording which required motifs have already appeared.

An insertion, deletion or mismatch resets the contiguous-prefix memory. It does not erase motifs found earlier. A terminal alignment is accepted only when every required motif has occurred. This semantics means a motif must appear in contiguous exact paired columns. Merely occurring in both unaligned source sequences is insufficient. Up to four supplied motifs of at most sixteen characters are supported in this bounded implementation.

The product graph retains exact feature multiplicities and supports selection within its constrained fibres. The tests independently render every small admissible alignment, split it into contiguous matching runs, and compare direct motif acceptance with the product-DAG population. This extension illustrates how PerfectPower can move from aggregate numerical constraints to regular-language constraints on complete paths. It creates a basis for later finite automata expressing ordered motifs, bounded interruptions or other precisely specified alignment requirements, without claiming that those additional languages are already implemented.

Importing BLAST evidence and preserving coordinates

The strict importer expects the twelve standard tabular fields together with BTOP; sequence lengths and taxonomy can be retained as additional metadata. It supports outfmt 6 with an explicit field order and outfmt 7 with a recognized Fields header. It verifies the relation among traceback length, identical residues, mismatches, gap openings and coordinate spans. Reported percentages are rounded decimal quantities, so their comparison with exact identity fractions uses the precision of the printed decimal rather than exact decimal equality.

E-values and bit scores are retained as finite nonnegative decimal strings. They are never converted to binary floating-point values for acceptance. An E-value of zero in a report is preserved as the reported value; it is not treated as a statement that a chance event has mathematical probability zero. Taxonomy metadata can be unavailable in pairwise local searches and is retained without inventing a taxon.

Traceback replay extracts the indicated intervals from supplied source records and checks each matching run, substitution and gap residue. Reverse nucleotide coordinates cause a reverse-complement extraction with explicit IUPAC complement rules. Reverse protein coordinates are rejected. Source record hashes accompany replay results, so agreement refers to specific supplied bytes and not an accession name alone.

The importer deliberately rejects translated coordinate spans that would require a codon-aware adapter. Protein BLAST tracebacks can be inspected in their residue coordinates, and scalar alignment can use an explicit exact substitution table covering every encountered residue pair. Composition-adjusted protein scoring, PSSM generation and translated six-frame interpretation are not implemented. The default feature polynomial uses literal identity and mismatch counts rather than pretending to encode a general protein substitution matrix.

Connecting fragments into coherent chains

A BLAST result can contain multiple HSPs for one query and subject. The chain population groups compatible fragments by query identifier, subject identifier and relative orientation. Both coordinate axes must increase after orienting them according to their own search directions. Fragments cannot overlap on either axis in the current chain model. Optional query and subject spacing limits bound the intervening unaligned distances.

An optional modulus checks a user-specified spacing relation between subject and query coordinates. This is an arithmetic constraint; it does not infer coding phase, establish a transcript model or substitute for annotated exon structure. Query coverage is the sum of nonoverlapping query interval lengths, which equals their union length under the model's nonoverlap rule. Scores add the individual fragments' alignment features. Inter-fragment spacing is not automatically charged as an alignment gap.

The compiler enumerates every nonempty admissible chain of the retained HSP graph, subject to an explicit work limit. It stores compatible edges and reasons for rejecting incompatible edges, including query overlap, subject ordering, spacing and frame-spacing violations. This makes exclusions inspectable. The result can help a researcher organize evidence for a repeated region or a possible rearrangement, but compatible fragments alone do not establish such a biological event.

Three transport semirings and all-future state equivalence

The alignment DAG also connects to the repository's existing typed Transport objects. In count mode, transition weights are one and composition counts paths. In rational mass mode, each edge contributes the product of its feature activities. In min-plus mode, edge costs are the negative of their raw-score increments, so minimizing complete-path cost maximizes raw score.

A path consumes at least one source character per nonterminal transition. A horizon equal to the sum of the source lengths is therefore sufficient. Terminal states receive an absorbing identity transition only in the transport calculation; a completed path is extended uniquely to that common horizon. This avoids counting one biological operation path once at every possible stopping time. The tests compare count, partition mass and minimum negative score with the separate polynomial family calculation.

The SOE adapter uses a different model: a finite deterministic partial machine whose action labels distinguish matches, mismatches, deletion opening, deletion extension, insertion opening and insertion extension. Terminal acceptance is its observation. Encoding feature increments in these labels retains more

information than using unlabelled generic moves. The existing generic SOE compiler proves observation transport, transition transport and separating experiments for quotient classes, and invokes its general all-future equivalence theorem.

The retained AC versus AG machine has thirteen states and eight quotient classes. Lean checks its literal model and quotient, including behavior under every finite action word. A separate motif-prefix machine receives the same treatment. This establishes the advertised semantics of the supplied finite tables. It does not prove that every Python-to-table adapter or every biological interpretation is correct, and it does not promise that every alignment model admits a large state reduction.

The new generic Lean path-length theorem shows that strictly decreasing natural ranks bound the length of every executable word. The retained alignment packet checks rank decrease and disabled actions at terminal states. Its bounded recursive accepting-word count is exactly three at the original state with horizon four. These checks connect the finite model's acyclicity, stopping semantics and retained path count without introducing additional mathematical axioms.

Polynomial activities, Gamma counts and inspection policies

A finite alignment family can be given positive rational activities for each feature. Evaluating its PSG polynomial gives an exact partition mass. Differentiating that polynomial gives exact feature means and covariance under the normalized activity ensemble. At all activities equal to one, the distribution weights operation paths uniformly. This is a declared mathematical ensemble, not an evolutionary posterior learned from real data.

The finite-jet adapter uses existing PSG convolution to compute Taylor coefficients along an affine activity path. Its order is explicit. Since the underlying generating object is a finite polynomial, these are exact coefficients; the routine does not fill unavailable higher-order data with zeros to claim a deeper analytic result. This interface makes parameter sensitivity inspectable using machinery already developed for other PerfectPower objects.

A separate fixed-comparison null calculation provides an exact binomial tail for a stated number of identical ungapped position pairs under an IID Bernoulli model. Its coefficients connect to the existing fixed-width Gamma/binomial normalization. The result is explicitly labelled as a fixed, unselected comparison. Alignment optimization, database search, gaps, sequence dependence and composition bias change the statistical problem, so this tail must not be substituted for a BLAST E-value.

The diagnostic adapter presents competing feature summaries as finite hypotheses to the repository's optimal resettable diagnostic-policy engine. A measurement reveals a feature coordinate at a supplied cost. The resulting decision tree minimizes worst-case cost under noiseless exact observations and the finite measurement menu. Identical feature vectors remain indistinguishable under this menu. These are optimal inspection policies for mathematical summaries, not prescriptions for laboratory experiments or diagnostic tests on patients.

Similarity networks, topology and strand transport

The network builder requires a qualifying individual HSP in both search directions before creating a conservative reciprocal edge. Identity, E-value and coverage thresholds are explicit. Coverage for this builder is per HSP, rather than a silently merged collection of fragments. Missing sequence lengths prevent a coverage-qualified edge. Conflicting strand signs remain unresolved and are excluded from the strand-labelled network instead of being arbitrarily resolved.

A supplied weighted network can then pass to the exact Hodge module. Its incidence boundary records oriented graph edges. Optionally, every complete triangle is filled as a two-dimensional face. The chain identity boundary¹ times boundary² equals zero is checked exactly. Positive rational mass matrices yield gradient, triangle-boundary and harmonic projectors, with their idempotence, orthogonality and Laplacian-kernel identities replayed algebraically.

The distinction between graph cycles and holes after filling triangles matters. A three-edge triangle has one graph cycle, but that cycle is a boundary when its triangular face is present. Both interpretations can be requested. The threshold filtration inserts equal-weight edges together and records exact component counts and graph cycle ranks at every distinct supplied weight. These are properties of the constructed evidence network, not automatic evidence of phylogenetic branching or gene transfer.

Strand signs also define a $\mathbb{Z}/2$ gain graph through the existing ConnectionGraph machinery. A cycle with inconsistent orientation product has nontrivial transport. This can expose contradictory strand relationships in a constructed network. It should prompt inspection of the source alignments and modelling choices rather than be interpreted as a Riemann surface or a discovery of genomic monodromy. Exact Laplacian trace moments offer numerical invariants without making floating-point eigenvalue decisions.

Acquiring inputs and reproducing the implementation

The NCBI client retrieves batches of versioned nucleotide or protein accession records through EFetch, verifies the returned record identifiers, and caches FASTA bytes with hashes. Remote BLAST submission requires an explicit query, database, program and contact email. It stores the request identifier and query hash. Retrieval is a separate operation with persistent pacing; the software does not block in a hidden polling loop.

The shared BLAST service requires requests at least ten seconds apart and polling of one request identifier no more frequently than once per minute. The client enforces these intervals within its cache. Its present pacing state is intended for one sequential client using that cache, rather than concurrent processes racing on the same directory. Large workloads should use standalone BLAST+ or appropriate cloud resources. There is no automatic download of the enormous unrestricted nucleotide or protein databases.

The local BLAST wrapper uses an installed blastn or blastp executable, explicit argument lists and bounded process timeouts. It produces custom tabular output with BTOP and a receipt recording version, invocation and input/output hashes. The retained example is reproducible without making a new network request because its public FASTA records and original BLAST output are committed.

Install the branch package with `python -m pip install .` and invoke `python -m perfectpower.blast_cli --help`. The align command takes literal sequences and saves a complete-family analysis. The pair command accepts one-record FASTA files for scalar local or global alignment. The ingest command imports custom

BLAST output and optionally replays source records. The report command turns an alignment or demonstration receipt into self-contained HTML. The fetch, blast, submit and retrieve commands expose acquisition explicitly.

Run make blast-test for the dedicated independent checks, make blast-atlas to rebuild the retained mathematical demonstration, and make blast-lean to compile the new Lean bridge and finite packets in a configured Lean environment. The interactive atlas ships inside the installed wheel as package data. It uses integer scoring for the live sliders over saved feature support. Screen coordinates for drawing exact rational polygons are floating-point display values; the saved vertices remain the authoritative mathematical result.

A bounded scaling experiment

The retained scaling trial increases the length of an illustrative cyclically shifted DNA pair from four to twenty characters. At length twenty, its complete global model contains 3,772,698,725 operation paths, represented by 1,245 feature summaries and 188,596 retained suffix terms. Polynomial construction took about 0.37 seconds in this single-machine run, and its optimal score agreed with the independent scalar dynamic program. This is a compressed exact counting calculation; the software did not render billions of paths. The result demonstrates the value of keeping a solution population in compressed form rather than enumerating every member. It establishes neither a BLAST speedup nor the same performance on arbitrary sequences or richer biological models.

Validation, scope and the next scientific step

The dedicated suite contains twenty-one tests with independent enumerators and adversarial input cases. It checks complete feature multiplicities, rank/select inversion, scalar global optimum agreement, local optimum agreement with an all-substring oracle, exact parameter boundaries, tie retention, robustness, constrained queries, three transport semirings, conserved-motif path constraints, literal traceback replay, reverse complements, network reciprocity, Hodge dimensions, motif-word counts, acquisition caching and offline-report encoding. Existing SOE, PSG and Gamma regressions were also run; two pre-existing kernel-dependent checked-SOE cases were skipped in their ordinary test invocation, while the new literal packets were compiled separately with Lean.

Six listed Lean modules compile in the source-bound kernel build, including the existing SOE theorem module and five new modules. The new theorem families cover additive scoring, affine score lines, supported winners, executable-word length bounds, literal SOE acceptance, finite accepting-word counts and retained score comparisons. Their reported axioms are limited to standard Lean axioms. The full repository release-verification target was not run for this extension, and no full formal proof of the Python polynomial dynamic program is claimed.

The implementation is an exploratory research layer. Its next meaningful biological evaluation is a curated sequence family with known alternatives and annotations, measuring whether parameter maps, motif constraints and coherent-chain exploration reveal consequential ambiguity that a single ranked BLAST result hides. Search sensitivity, useful prioritization, reproducibility and human interpretability should be assessed separately from wall-clock performance. This release establishes a substantial mathematical and software foundation for that evaluation; it does not establish a novel biological discovery, an industrial speedup or a clinically validated application.

Primary sources and retained public records

NCBI BLAST Command Line Applications User Manual: <https://www.ncbi.nlm.nih.gov/books/NBK279690/> . Its configurable output, BTOP and affine-gap documentation define the interoperability target. NCBI BLAST developer guidance: <https://blast.ncbi.nlm.nih.gov/doc/blast-help/developerinfo.html> . NCBI Common URL API: <https://blast.ncbi.nlm.nih.gov/doc/blast-help/urlapi.html> . NCBI statistics tutorial: <https://www.ncbi.nlm.nih.gov/BLAST/tutorial/> .

Pachter and Sturmfels, Parametric inference for biological sequence analysis, PNAS 2004: <https://pmc.ncbi.nlm.nih.gov/articles/PMC528961/> . Dewey, Huggins, Woods, Sturmfels and Pachter, Parametric Alignment of Drosophila Genomes, PLOS Computational Biology 2006: <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.0020073> . These are established foundations for generating-polynomial and alignment-polytope interpretations. This extension's contribution is its integration with PerfectPower's existing machinery and explicit executable interfaces; it does not claim priority for those mathematical foundations.

Human transcript record: https://www.ncbi.nlm.nih.gov/nuccore/NM_000207.3 . Dromiciops gliroides transcript record: https://www.ncbi.nlm.nih.gov/nuccore/XM_043971863.1 . NCBI workshop scoring example using this transcript pair: https://www.nlm.nih.gov/ncbi/workshops/2023-08_BLAST_evol/appendix.html . Versioned FASTA files, their hashes, the retained BLAST invocation and mathematical receipts accompany the branch.